Syllabus Edition
First teaching 2025
First exams 2027

Not your syllabus?

Methods of Sustainable Production

Exam code: 0450, 0986 & 0264, 0774

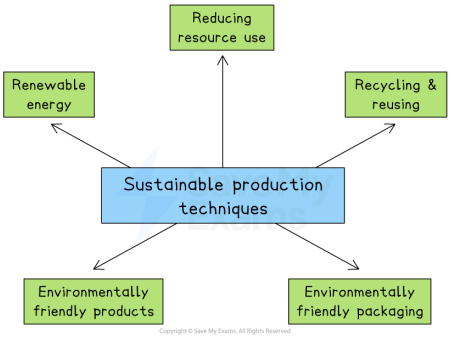
Download PDF

Written by: Lisa Eades
Reviewed by: Steve Vorster
Updated on 28 August 2025

Introduction to sustainable production

- **Sustainable production occurs when a business produces goods and services in ways that do not harm the environment, society or future generations**
  - It focuses on meeting current needs without using up natural resources or causing long-term damage

What sustainable production may involve



Sustainable production may involve using renewable energy, reducing resource use, recycling and reusing resources and developing environmentally friendly products or packaging

- These practices help **protect the environment**, improve a business's **reputation** and may also **reduce costs** in the long term

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Renewable energy comes from natural sources that will not run out, like the **sun, wind, water** and **geothermal heat**
- These sources are constantly being renewed by nature

How businesses use renewable energy

- **Solar panels** on buildings to generate electricity
- **Wind turbines** to produce power, especially for factories or large offices
- **Hydroelectric power,** if the business is located near rivers or dams
- **Geothermal heating** for buildings in suitable areas
- **Buying green energy** from suppliers who use renewable sources

Case Study

Renewable Energy Usage at IKEA

Global furniture retailer IKEA has invested heavily in renewable energy

- Installed **solar panels** on many store rooftops
- Built **wind farms** in countries including Germany and the USA
- Committed to producing more energy than they use by 2025

Benefits

- **Lower energy costs** for stores and warehouses
- Improved **brand reputation** as a sustainable company
- Attracted **environmentally conscious customers**
- Helped IKEA reach its goal of being climate positive

Help me understand

Reducing resource use

- Reducing resource use means using **fewer materials, water** and **energy** when making goods or providing services

ways:

- **Designing products** that use fewer materials
- **Reusing materials** in production instead of using new ones
- **Improving efficiency**, like using machines that need less electricity
- **Reducing packaging**, or switching to recycled or recyclable packaging
- **Saving water**, for example by using water-efficient systems

Reusing and recycling

- **Reusing** means using items or materials again instead of throwing them away
- **Recycling** is turning waste materials (like paper, plastic or metal) into new products
- Businesses reuse and recycle in the following ways:
  - **Reuse packaging**, containers, or leftover materials in production
  - **Recycle waste**, like paper, plastic, glass, or metal, either in-house or through recycling companies
  - **Encourage customers** to return used products (e.g. refillable bottles or old electronics)
  - **Buy recycled materials** to make new products

Case Study

Reusing and Recycling at Patagonia

Patagonia, a clothing company, is well-known for its strong environmental focus

- **Repairing clothes** for customers instead of encouraging new purchases
- Using **recycled materials** like plastic bottles and old clothes to make new jackets
- Running its **Worn Wear** programme, where customers return used clothes for reuse or resale

Benefits

- It has a strong **brand image** for caring about the environment
- It has developed strong **customer loyalty**, especially from environmentally conscious buyers

Developing environmentally friendly products

- Environmentally friendly products are designed to **have less harm on the planet**
  - They use **natural** or **recycled materials**
  - They are **energy efficient**
  - They create **less waste** or **pollution**
  - They are **biodegradable** or **reusable**
- These products help reduce environmental impact during use and after disposal
- Businesses develop environmentally friendly products in the following ways:
  - Use **sustainable materials**, like organic cotton or recycled plastic
  - Design products that **last longer** or are **easy to repair**
  - Replace harmful chemicals with **natural alternatives**
  - Offer **refillable** or **reusable** versions of existing products

Case Study

Lego's Environmentally Friendly Products

Toy company LEGO has replaced replacing oil-based plastic with **plant-based plastic** made from sugarcane

- Launched eco-friendly **LEGO plant pieces** (like trees and bushes)
- Invested in research to develop **fully sustainable building bricks**

Benefits

- Helped reduce use of **fossil fuels**
- Improved LEGO's **brand image** as an eco-conscious company
- Gained **positive media coverage** and support from parents and schools
- Helped LEGO work towards its goal of making all products sustainable by 2030

Using environmentally friendly packaging

- Environmentally friendly packaging is made from materials that are **recycled, recyclable, biodegradable, compostable** or **reusable**
- It is designed to **reduce waste** and **lower harm to the environment**
- **B**usinesses develop environmentally friendly packaging in the following ways:
  - Use **paper or cardboard** instead of plastic
  - Choose **biodegradable** or **compostable bags**
  - Offer **refillable packaging** or reusable containers
  - Use **minimal packaging** to avoid waste
  - Print with **eco-friendly inks** or labels

Case Study

Environmentally Friendly Packaging at McDonalds

McDonald's has made major changes to its packaging in many countries in recent years

- Replaced **plastic straws** with paper ones
- Switched to **recyclable paper wrappers and cups**
- Stopped using **foam packaging**
- Started using **certified sustainable materials** in paper products

Benefits

- **Less plastic waste**, especially in oceans
- **Improved reputation**, especially with younger customers
- Helped meet government rules on reducing plastic
- Improved McDonald's **competitiveness** with other eco food brands

Examiner Tips and Tricks

Use precise examples when explaining sustainability, such as recycling materials or switching to renewable energy. Concrete methods show stronger understanding than vague statements

Advantages and disadvantages of sustainable production

- In 2024, 91% of new renewable energy projects were cheaper than fossil fuels
  - Solar was 41% cheaper and onshore wind 53% cheaper on average
  - These savings helped global businesses save up to **$467 billion in fuel costs**
- However, sustainable businesses often have a lot of extra work, like checking records and collecting data to make sure they pass regular audits

Evaluation of sustainable production

Advantages	Disadvantages
• Sustainable production improves the business **image** and attracts more customers • It can **reduce long-term costs** by using less energy and fewer materials	• Sustainable production can be **expensive** to start using new equipment or processes • Some environmentally friendly materials may be more expensive than traditional ones

• It encourages **innovation** and can make the business more efficient	methods • Changing how products are made can **take time** and cause short-term delays

Greenwashing

- Greenwashing is when a business **pretends to be environmentally friendly**, but its actions do not match what it claims
  - For example, a company might advertise its products as eco-friendly when, in reality, they are not
- It is a way for businesses to **attract customers** who care about the environment, **without making real changes** to how they work

Problems of greenwashing

- **Loss of trust**
  - If customers find out the business lied, they may stop buying from it
  - Eco-conscious customers may share negative reviews or avoid the brand entirely
  - It can lead to bad publicity and complaints
- **Legal trouble**
  - In some countries, making false environmental claims can lead to fines or court action

Teacher only

New! Exclusive Teacher-Only Questions

Fresh, expert-written questions

Students can't access them

Fair accurate tests - built fast

Explore in Test Builder

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.